

Ritesh Raj

APPLIED AI ENGINEER

rajritesh03@gmail.com | 0415745143 | ritesh-raj-portfolio.vercel.app/ | github.com/ritesh28 | linkedin.com/in/raj-ritesh

SUMMARY

Applied AI Engineer specializing in building and deploying production-grade AI systems for compliance, risk, and assurance applications. Hands-on experience developing LLM applications, multi-agent workflows, semantic retrieval systems, AI orchestration, sandboxed agent execution, and agentic loops. Designed and deployed AI capabilities supporting **100+ users** across **1,000+ documents**, **reducing manual compliance review effort by 60%**. Strong software engineering foundation across Python, TypeScript, Azure, and modern AI application frameworks, with a focus on reliable, scalable, and production-ready AI solutions.

WORK EXPERIENCE

PwC — Senior Analyst

May 2022 – Present

- Designed and built **Pathfinder**, a production **AI-powered compliance platform** that analyzes regulations against organizational policies, controls, standards, and other artefacts to identify **compliance gaps** and areas requiring review.
- Delivered **production AI capabilities** used by **100+ users** across **1,000+ documents**, reducing manual compliance review effort by **60%**.
- Designed a scalable **multi-agent architecture** and **orchestration layer** for client-specific compliance and risk assessment workflows, enabling organizations to apply their own policies, controls, methodologies, and decision criteria without separate implementations.
- Built an **agentic matter-triage workflow** using **Daytona sandboxes** and **Python-based extraction** to process Excel-based matter lists, normalize matters into a standard schema, and prepare structured data for downstream AI workflows.
- Designed an **orchestrator agent** that dynamically spawns specialized agents for **requirement checks, incident definition, impact assessment, threshold validation, and root-cause analysis**, enabling automated multi-step assessment workflows.
- Implemented **semantic retrieval** using **embeddings in PostgreSQL** and designed a **vectorless RAG architecture** using structured **Markdown (.md) skills and subskills** for targeted knowledge retrieval.
- Developed a **Node.js ingestion and tool-calling pipeline** that parses skill and subskill Markdown files and dynamically exposes them as **callable AI tools**, enabling flexible **skill discovery and agent orchestration**.
- Integrated multiple **LLM providers** through an enterprise AI platform using a **unified model interface**, reducing application-level coupling to individual model providers.
- Built **human-in-the-loop** calibration and **agentic feedback loops** that incorporate user corrections into downstream workflows, enabling validated decisions to be applied consistently across similar assessments.
- Developed an **LLM evaluation framework** covering **scope adherence, semantic keyword coverage, and task completion**, and established a **feedback loop** using Linear to drive continuous improvement of prompts and AI workflows.
- Implemented **AI observability and traceability** using **Langfuse and ClickHouse** for LLM cost monitoring, workflow tracing, and debugging complex **multi-step agent execution**. Maintained production systems using **TypeScript, React, Next.js, NestJS, GraphQL, PostgreSQL, Docker, Kubernetes, and RabbitMQ**.

Infosys — Technology Analyst

August 2015 – May 2022

- Developed **Information Document Analyzer (IDA)**, an **AI-driven document processing solution** that reduced manual data-entry effort by **70%** through automated data extraction and field inference.
- Built an **intelligent document extraction pipeline** using **Azure Form Recognizer**, enabling users to annotate key-value fields on sample documents and train custom extraction models for automated processing.
- Designed an end-to-end **document AI workflow** covering document upload, manual field annotation, **model training, inference, and automated key-value extraction** across newly uploaded documents.
- Implemented **event-driven microservices** using **Python and RabbitMQ** to orchestrate document processing and model inference workflows at scale.
- Developed the **React.js** interface for document upload, field annotation, and extraction workflows, integrated with **Azure AI services** and Python-based backend microservices.
- Developed **Customer Consent Management** capabilities for **Open Banking**, enabling secure data sharing with accredited third parties for **100K+ users**. Built reusable **React.js** components, **Node.js REST APIs, JWT authentication, Redis**

caching, and **Splunk** monitoring.

- Built a **Test Environment Management System (TEMS)**, accelerating release cycles by **20–25%** and improving productivity by **30–40%** through early identification of configuration conflicts. Developed **React.js** interfaces, integrated **Jira APIs**, and implemented **Node.js** backend services with **SQL Server**.

PERSONAL PROJECTS

VFS Marketplace Agent

Demo app that gives an AI agent a **virtual file system (VFS)** as its working surface for a small marketplace domain (sellers, customers, support).

- GitHub: <https://github.com/ritesh28/vfs-marketplace-agent>
- Demo: <https://vfs-marketplace-agent.vercel.app/>
- Built a **persona-scoped VFS** with lazy/JIT hydration so the agent explores a closed file tree instead of loading full marketplace state into the prompt.
- Implemented VFS tools (`list_directory` , `read` , `write` , `search`) where the agent reads **markdown projections** of DB records and writes structured **JSON** under `agent-output/` ; heuristic mapping updates **Postgres**, appends an **events** audit row, and refreshes domain `.md` files.
- Enforced **closed path allowlists**, user/path guardrails, and create-only writes validated with **Zod**, keeping Postgres as the source of truth.
- Shipped a **Next.js** UI with streaming **multi-provider** chat (**OpenAI**, **Gemini**, **Anthropic** via AI SDK), live VFS mirror, and Database tab; stack includes **TypeScript**, **Drizzle**, **Postgres**, and **PostHog**.

Skill Tree RAG

Terminal chatbot where **skills** (markdown trees) are first-class tools for on-demand agent tool calling.

- GitHub: <https://github.com/ritesh28/skill-tree-rag>
- Built a **skill-sync** pipeline that turns markdown skill trees under `skills/` into generated TypeScript tool definitions for agent tool calling.
- Added skill structure, reference, and frontmatter checks plus **Husky** pre-commit hooks so invalid or unreferenced skills fail before commit.
- Supports multiple LLM providers (API keys or local **Ollama**), streamed TUI chat with interrupt, and clarifying `ask_question` prompts; optional **Python/uv** skill scripts for PDF create/split/read via **pymupdf**.

SKILLS

- **Applied AI**: LLM Applications, Multi-Agent Systems, Agentic Workflows, AI Orchestration, Tool Calling, RAG, Vectorless RAG, Semantic Retrieval, Embeddings, Prompt Engineering, Human-in-the-Loop AI, Agentic Loops
- **AI Engineering**: Python, LangChain, Vercel AI SDK, scikit-learn, TensorFlow, NumPy, Pandas, NLTK, Matplotlib, Document AI, Information Extraction, Azure Form Recognizer
- **AI Evaluation & Observability**: LLM Evaluation, Langfuse, ClickHouse, Linear
- **Infrastructure & Workflows**: Sandboxed Agent Execution, LangGraph, Azure, Docker, Kubernetes
- **Software Engineering**: TypeScript, NestJS, Next.js, GraphQL, PostgreSQL, RabbitMQ

EDUCATION

Bachelor in Computer Engineering, University of Mumbai -2015